Low storage and efficient 3D reverse time migration method based on eikonal equation

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Abstract

In 3D reverse time migration (RTM), high computational costs and massive storage requirements remain major challenges that are largely determined by the different imaging conditions. The cross-correlation imaging condition (CCIC) provides high imaging resolution but requires storing a large volume of source-wavefield data on the disk and subsequently reading these data back into the memory during imaging. Consequently, the substantial storage demand, intensive input/output operations, and limited data transfer bandwidth substantially reduce computational efficiency. In contrast, the excitation amplitude imaging condition (EAIC) alleviates these limitations by determining the imaging time through the identification of the maximum amplitude of the source wavefield at every grid. However, peak amplitude selection becomes unstable in complex media, producing discontinuous imaging times and distorted excitation amplitudes, leading to reduced imaging accuracy and poor noise robustness. To address this challenge, we propose a novel method for computing the imaging time by solving the 3D eikonal equation using an adaptive finite-difference scheme that adaptively selects appropriate local operators according to the dominant wave propagation modes, thereby providing more accurate traveltimes for imaging time estimation. The proposed method retains the low storage requirement and high computational efficiency of the EAIC while significantly enhancing the resolution of the migration images. Numerical experiments on 3D

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simple and complex Salt models demonstrate that our proposed method achieves imaging accuracy comparable to that of CCIC while avoiding the computational bottlenecks associated with massive wavefield storage and data transfer. Moreover, it effectively overcomes the limited imaging accuracy and poor noise robustness of the conventional EAIC, thereby showing considerable potential for practical applications.

Keywords**:** 3D reverse time migration; excitation amplitude imaging condition; 3D eikonal equation; first-arrival traveltimes; adaptive finite-difference method

1. Introduction

To meet the growing demand for petroleum extraction, the industry requires higher seismic imaging accuracy and lower computational costs. The quality of migration directly affects the precision of reservoir characterization. Pre-stack depth migration, as the mainstream imaging method in seismic exploration, can be broadly divided into ray-theory-based methods, including Kirchhoff migration (Schneider 1978; Chang *et al.* 1998; Pasasa *et al.* 1998; Yue *et al.* 2023) and Gaussian beam migration (Hill 2001; Gray 2005; Zhu *et al.* 2007; Hu *et al.* 2020), and wave-equation-based methods, including one-way wave equation migration (Gazdag & Sguazzero 1984; Mulder & Plessix 2004; Zhang *et al.* 2005) and reverse time migration (RTM) (Baysal *et al.* 1983; Loewenthal & Mufti 1983). Currently, RTM stands out among pre-stack depth migration methods because of its ability to handle two-way wave equations and its lack of non-dip angle limitations, making it one of the most advanced imaging tools (Du *et al.* 2014; Zhou *et al.* 2018).

RTM is generally implemented in three key stages: forward propagating a flat wavelet to obtain the source wavefield, backward propagating the receiver seismic data, and selecting an appropriate imaging condition to obtain the final image. Among these steps, the imaging condition plays a pivotal role because it directly influences both the efficiency of the imaging calculation and the resolution of the resulting images.

Claerbout (1971) proposed a famous zero-lag cross-correlation imaging condition (CCIC), that

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calculates the migration image by correlating the source wavefield with the receiver wavefield at each time step. However, the images produced using this imaging condition lack a physical interpretation of the true reflection coefficients. To overcome this constraint, Kaelin and Guitton (2006) proposed a source-normalized cross-correlation imaging condition (SNCCIC) as an improved approach. This method normalizes the cross-correlated image by dividing it by either the squared source or receiver illumination intensity. The SNCCIC effectively eliminates low-frequency artifacts while restoring the true reflection coefficients (Chattopadhyay & McMechan 2008). However, the application of SNCCIC in 3D scenarios is challenging, as it requires saving all source or receiver wavefield data to the disk and subsequently reading them to memory for imaging calculations. The resulting storage demand and frequent data exchanges among the disk, CPU, and GPU substantially reduce RTM efficiency (Zhang *et al.* 2019).

Several source-wavefield reconstruction strategies have been developed to reduce storage. Boundary storage methods (Feng & Wang 2012; Nguyen & McMechan 2015) retain wavefields within boundary grid layers and use them as boundary conditions during backward propagation. However, memory demand increases with the order of the finite-difference stencil and remains substantial for high-order 3D simulations. Random boundary reconstruction (Clapp 2009) requires no additional storage and can reconstruct any wavefield in only two recursive steps; however, the random boundary condition introduces random noise into the migration images. Another type of wavefield reconstruction method is the checkpointing method (Griewank 1992; Symes 2007), which stores the wavefield at selected time intervals and recomputes the source wavefield between the checkpoints. Although the checkpointing method alleviates storage demands, its high recomputation rate results in a low computational efficiency (Chen & Wang 2018).

The excitation amplitude imaging condition (EAIC) (Nguyen & McMechan 2013) provides a low-storage alternative to SNCCIC. The EAIC determines the imaging time by identifying the peak amplitude value at every grid and uses only the wavefield information at the imaging time. This

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strategy significantly reduces the storage and I/O; however, peak amplitude picking can be unstable in complex media and does not account for multipathing, which may distort the final image. Incorporating first-arrival traveltimes into the EAIC has been demonstrated to effectively improve imaging quality (Gu *et al.* 2015; Li *et al.* 2024; Wang *et al.* 2025). Many researchers have explored methods for solving the eikonal equation to efficiently and accurately calculate first-arrival traveltimes (Sethian 1996; Zhao 2005; Zhang *et al.* 2006; Fomel *et al.* 2009; Noble *et al.* 2014; Zhang *et al.* 2024). However, owing to the source singularity, significant errors occur in the first-arrival traveltimes near the source location, and these errors accumulate as the propagation distance increases. Because near-offset seismic data are crucial for seismic imaging, Qiao *et al.* (2021) proposed an adaptive finite-difference method (AFDM) to obtain more accurate first-arrival traveltimes and extended the approach to 3D applications (Qiao *et al.* 2024). This method adaptively determines the most suitable local operators through a combined application of Fermat's principle and the global fast sweeping method, thereby achieving enhanced first-arrival traveltime accuracy.

In this study, we solved the 3D eikonal equation using AFDM and used the resulting traveltimes as a reference to determine the imaging time, which enables the extraction of continuous and clear excitation amplitudes, thereby improving the resolution of the migration results. The rest of this paper is organized as follows: Section II reviews 3D RTM and AFDM operators and our proposed improved EAIC. Section III presents comprehensive tests and analyses of our method using both 3D simple and Salt models. Section IV discusses the limitations of the proposed approach and potential improvement strategies. Finally, Section V concludes the paper with a summary of key findings.

2. Theory

2.1. The review of RTM

We modeled the seismic wave propagation using the second-order acoustic velocity-stress wave equation

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$$\frac{1}{v(\mathbf{x})^2}\frac{\partial^2 P(\mathbf{x},t)}{\partial t^2}=\frac{\partial^2 P(\mathbf{x},t)}{\partial x^2}+\frac{\partial^2 P(\mathbf{x},t)}{\partial y^2}+\frac{\partial^2 P(\mathbf{x},t)}{\partial z^2}, \tag{1}$$

where $\mathbf{x} = (x, y, z)$ represents the 3D spatial coordinates, v is the velocity of the subsurface media, P is the pressure wavefield, and t is time. By incorporating the source wavelet into Equation (1), we simulated the source wave propagation via

$$\begin{cases}\dfrac{1}{v^2}\dfrac{\partial^2 P_{\mathrm{s}}(\mathbf{x},t)}{\partial t^2}=\dfrac{\partial^2 P_{\mathrm{s}}(\mathbf{x},t)}{\partial x^2}+\dfrac{\partial^2 P_{\mathrm{s}}(\mathbf{x},t)}{\partial y^2}+\dfrac{\partial^2 P_{\mathrm{s}}(\mathbf{x},t)}{\partial z^2},\\ P_{\mathrm{s}}(\mathbf{x},t)=P_{\mathrm{s}}(\mathbf{x},t)+\delta(\mathbf{x}-\mathbf{x}_{\mathrm{s}})f(t),\end{cases} \tag{2}$$

and simulated the receiver wavefield by incorporating the seismic data at the receiver location into Equation (1) via

$$\begin{cases}\dfrac{1}{v^2}\dfrac{\partial^2 P_{\mathrm{r}}(\mathbf{x},t)}{\partial t^2}=\dfrac{\partial^2 P_{\mathrm{r}}(\mathbf{x},t)}{\partial x^2}+\dfrac{\partial^2 P_{\mathrm{r}}(\mathbf{x},t)}{\partial y^2}+\dfrac{\partial^2 P_{\mathrm{r}}(\mathbf{x},t)}{\partial z^2},\\ P_{\mathrm{r}}(\mathbf{x}-\mathbf{x}_{\mathrm{r}};t)=u(\mathbf{x}_{\mathrm{r}};t),\end{cases} \tag{3}$$

where P_{s} and P_{r} are the source and receiver wavefields, respectively, $f(t)$ represents the source wavelet function, $\mathbf{x}_{\mathrm{r}}$ denotes the coordinates of the receiver, and $u(\mathbf{x}_{\mathrm{r}}; t)$ represents the seismic records.

In this study, we solved Equations (2) and (3) using the 16th high-order finite-difference scheme, and the convolutional perfectly matched layer (CPML) absorbing boundary conditions were employed to suppress artificial boundary reflections. Furthermore, a collaborative CPU/GPU implementation improves the computational efficiency.

The final step of RTM is to output the migration image by applying the imaging condition that correlates the corresponding source and receiver wavefields. The CCIC is

$$I_{\mathrm{CCIC}}(\mathbf{x};s_j)=\sum_t P_{\mathrm{s}}(\mathbf{x};t;s_j)P_{\mathrm{r}}(\mathbf{x};t;s_j), \tag{4}$$

where s_j represents the jth source, and I_{CCIC} is the corresponding migration image. Compared with CCIC, SNCCIC preserves the true subsurface reflection coefficients while providing better

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illumination compensation for deep structures, as expressed by

$$I_{\mathrm{SNCCIC}}(\mathbf{x};s_j)=\frac{\sum_t P_{\mathrm{s}}(\mathbf{x};t;s_j)P_{\mathrm{r}}(\mathbf{x};t;s_j)}{\sum_t P_{\mathrm{s}}^2(\mathbf{x};t;s_j)}. \tag{5}$$

Equations (4) and (5) require the wavefield information at every time step, leading to expensive storage and computational costs, particularly in 3D cases. EAIC avoids this requirement by evaluating the receiver wavefield only at the excitation time, and the corresponding imaging condition is given by

$$I_{\mathrm{EAIC}}(\mathbf{x};s_j)=\frac{R(\mathbf{x};t_{\mathrm{e}};s_j)}{S_{\max}(\mathbf{x};t_{\mathrm{e}};s_j)}, \tag{6}$$

where $S_{\max}$ is the excitation amplitude, which is the maximum value in all grids across all time steps, t_{e} denotes the excitation (imaging) time, and R represents the receiver amplitude value at the corresponding imaging time. Equation (6) provides an optimal solution to this challenge. However, in complex geological environments, the imaging time and its corresponding amplitude calculated using EAIC become disordered, leading to distorted migration images. We therefore propose computing more reliable imaging time from the 3D eikonal equation using AFDM. This modification preserves the storage and efficiency advantages of EAIC while improving the image quality.

2.2. The review of AFDM

The eikonal equation can be formulated as

$$\begin{cases} |\nabla T(\mathbf{x})| = S(\mathbf{x}), & x\in\Omega, \\ T(\mathbf{x})=0, & x\in\partial\Omega, \end{cases} \tag{7}$$

where T represents the first-arrival traveltimes, S denotes the slowness, $\Omega \in \mathbb{R}^n$ indicates an n-dimensional spatial domain, and ∇ symbolizes the Hamiltonian operator.

Figure 1 shows a representative grid cell from the discretized slowness model to illustrate the local operators used to solve the 3D eikonal equation in Equation (7). The grid cell represents slowness, whereas the grid vertices represent first-arrival traveltimes. The AFDM selects different

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local operators to simulate plane waves or spherical waves propagating through the grid cell (green arrow), plane waves or spherical waves propagating along the grid-cell faces (red arrows), and direct or refracted waves propagating along the edges of the grid cells (blue arrows) to determine the first-arrival traveltime. The five local solution operators are the plane wave eight-point (P8), plane wave four-point (P4), spherical wave two-point (S2), spherical wave four-point (S4), and refracted wave two-point (R2) operators.

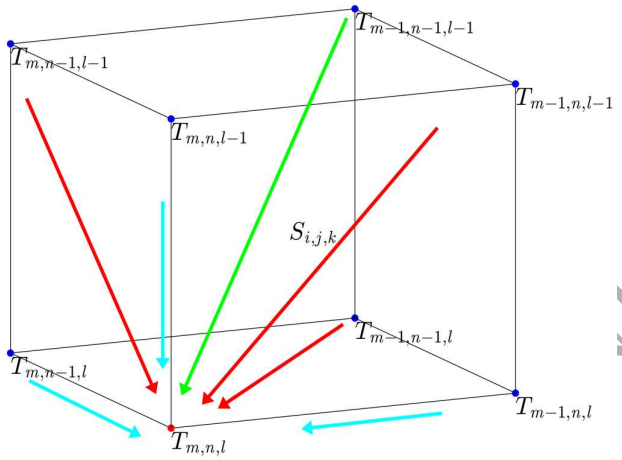


Figure 1 Schematic 3D distribution of traveltimes and slowness. Grid representation showing traveltimes stored at vertices and slowness defined at cell centers. Green arrow indicates plane or spherical wave propagating through grid cells, red arrows indicate plane or spherical waves propagating along grid-cell faces, and blue arrows indicate direct or refracted waves propagating along grid edges.

The P8 operator enables an accurate traveltime calculation for plane waves propagating through grid cells (see green arrows in Fig. 1). The spatial derivatives in Equation (7) are approximated using central finite differences as

$$\begin{cases} \dfrac{\partial T}{\partial x} = \dfrac{T_{m,n,l} - T_{px}}{4dx}, \\ \dfrac{\partial T}{\partial y} = \dfrac{T_{m,n,l} - T_{py}}{4dy}, \\ \dfrac{\partial T}{\partial z} = \dfrac{T_{m,n,l} - T_{pz}}{4dz}. \end{cases} \tag{8}$$

For simplicity, we define the intermediate variables T_{px}, T_{py} and T_{pz} as

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$$\begin{cases} T_{px} = T_{m-1,n,l} - T_{m,n-1,l} + T_{m-1,n-1,l} - T_{m,n,l-1} + T_{m-1,n,l-1} - T_{m,n-1,l-1} + T_{m-1,n-1,l-1}, \\ T_{py} = T_{m,n-1,l} - T_{m-1,n,l} + T_{m-1,n-1,l} - T_{m,n,l-1} + T_{m,n-1,l-1} - T_{m-1,n,l-1} + T_{m-1,n-1,l-1}, \\ T_{pz} = T_{m,n,l-1} - T_{m-1,n,l} + T_{m-1,n,l-1} - T_{m,n-1,l} + T_{m,n-1,l-1} - T_{m-1,n-1,l} + T_{m-1,n-1,l-1}. \end{cases} \tag{9}$$

Substituting Equations (8) and (9) into Equation (7) gives

$$\left(\frac{T_{m,n,l} - T_{px}}{4dx}\right)^2 + \left(\frac{T_{m,n,l} - T_{py}}{4dy}\right)^2 + \left(\frac{T_{m,n,l} - T_{pz}}{4dz}\right)^2 = S_{m,n,l}^2. \tag{10}$$

The solution for $T_{m,n,l}$ is obtained from the quadratic formula as

$$\begin{cases} T_{m,n,l} = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \\ a = \dfrac{1}{16}\left(\dfrac{1}{dx^2} + \dfrac{1}{dy^2} + \dfrac{1}{dz^2}\right), \\ b = -\dfrac{1}{8}\left(\dfrac{T_{px}}{dx^2} + \dfrac{T_{py}}{dy^2} + \dfrac{T_{pz}}{dz^2}\right), \\ c = \dfrac{1}{16}\left(\dfrac{T_{px}^2}{dx^2} + \dfrac{T_{py}^2}{dy^2} + \dfrac{T_{pz}^2}{dz^2}\right) - S_{m,n,l}^2. \end{cases} \tag{11}$$

To satisfy the illumination condition, the quadratic discriminant (b^2–$4ac$) must be non-negative. A larger root is selected to satisfy causality. Therefore, the P8 operator updates $T_{m,n,l}$ from the seven neighboring nodes for plane-wave propagation along the green arrow in Fig. 1.

For plane waves propagating along the grid-cell faces (red arrows in Fig. 1), we employ the P4 operator to calculate traveltimes. Taking the *XY* face at $z = k$ as an example, unlike the P8 operator, the P4 operator retains only the spatial derivatives in the x and y directions, and Equation (7) becomes

$$\begin{cases} \dfrac{\partial T}{\partial x} = \dfrac{T_{m,n,l} - T'_{px}}{2dx}, \\ \dfrac{\partial T}{\partial y} = \dfrac{T_{m,n,l} - T'_{py}}{2dy}. \end{cases} \tag{12}$$

Similarly, we define the two intermediate variables T'_{px} and T'_{py} as

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$$\begin{cases} T'_{px} = T_{m-1,n,l} - T_{m,n-1,l} + T_{m-1,n-1,l}, \\ T'_{py} = T_{m,n-1,l} - T_{m-1,n,l} + T_{m-1,n-1,l}. \end{cases} \tag{13}$$

Substituting Equations (12) and (13) into Equation (7) gives

$$\begin{cases} (\frac{T_{m,n,l} - T'_{px}}{2dx})^2 + (\frac{T_{m,n,l} - T'_{py}}{2dy})^2 = S_m^2, \\ S_m = \min(S_{m,n,l}, S_{m,n,l+1}). \end{cases} \tag{14}$$

Similarly, we also employ the quadratic formula in Equation (14), and the solution of $T_{m,n,l}$ can be written as

$$\begin{cases} T_{m,n,l} = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \\ a = \dfrac{1}{4}(\dfrac{1}{dx^2} + \dfrac{1}{dy^2}), \\ b = -\dfrac{1}{2}(\dfrac{T'_{px}}{dx^2} + \dfrac{T'_{py}}{dy^2}), \\ c = \dfrac{1}{4}(\dfrac{T'^{\,2}_{px}}{dx^2} + \dfrac{T'^{\,2}_{py}}{dy^2}) - S_m^2. \end{cases} \tag{15}$$

For the *XZ* and *YZ* planes, $T_{m,n,l}$ can be derived in the same manner. By comparing the first-arrival traveltimes computed on the three planes, the minimum value is selected as the numerical travel-time solution at (*m*, *n*, *l*).

Because of the source singularity, the direct application of the P8 and P4 operators produces large near-source errors that contaminate the solution at greater propagation distances. Therefore, we employ the factored method (Fomel *et al.* 2009) and introduce S2 and S4 operators. The S2 operator aims to simulate the spherical wave propagation through the grid cells (green arrow in Fig. 1). Traveltime and slowness are factored as

$$\begin{cases} T(\mathbf{x}) = T_0(\mathbf{x})\tau(\mathbf{x}), \\ S(\mathbf{x}) = S_0(\mathbf{x})\alpha(\mathbf{x}), \end{cases} \tag{16}$$

where S_0 represents the slowness value of a homogeneous model at the source, and T_0 denotes the analytical traveltimes. Substituting Equation (16) into Equation (7) yields

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$$T0\left|\nabla\tau\right|^2+2T_0\tau\nabla T_0\bullet\nabla\tau+(\tau^2-\alpha^2)S_0^2=0, \tag{17}$$

with the boundary conditions

$$\tau(\mathbf{x})=\begin{cases}T(\mathbf{x})/T_0(\mathbf{x}), T_0(\mathbf{x})\neq 0,\\ \alpha(\mathbf{x}), T_0(\mathbf{x})=0.\end{cases} \tag{18}$$

The derivatives of τ in Equation (17) are discretized into three spatial dimensions as

$$\begin{cases}\dfrac{\partial\tau}{\partial x}=\dfrac{\tau_{m,n,l}-\tau_{m-1,n,l}}{dx},\\ \dfrac{\partial\tau}{\partial y}=\dfrac{\tau_{m,n,l}-\tau_{m,n-1,l}}{dy},\\ \dfrac{\partial\tau}{\partial z}=\dfrac{\tau_{m,n,l}-\tau_{m,n,l-1}}{dz},\end{cases} \tag{19}$$

Substituting Equation (19) into Equation (17), the solution for τ can be given as

$$\begin{cases}\tau_{m,n,l}=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a},\\ a=T_0^2(\dfrac{1}{dx^2}+\dfrac{1}{dy^2}+\dfrac{1}{dz^2})+2T_0(\dfrac{T_{0,x}}{dx}+\dfrac{T_{0,y}}{dy}+\dfrac{T_{0,z}}{dz})+S_0^2,\\ b=-2T_0^2(\dfrac{\tau_{m-1.n.l}}{dx^2}+\dfrac{\tau_{m,n-1,l}}{dy^2}+\dfrac{\tau_{m,n,l-1}}{dz^2})-2T_0(\dfrac{\tau_{m-1.n.l}T_{0,x}}{dx^2}+\dfrac{\tau_{m.n-1.l}T_{0,y}}{dy^2}+\dfrac{\tau_{m.n.l-1}T_{0,z}}{dz^2}),\\ c=T_0^2(\dfrac{\tau_{m-1.n.m}^2}{dx^2}+\dfrac{\tau_{m.n-1.l}^2}{dy^2}+\dfrac{\tau_{m.n.l-1}^2}{dz^2})-S_{m,n,l}^2.\end{cases} \tag{20}$$

The numerical solution for the traveltimes at each grid point can be calculated by combining Equations (16) and (20). Therefore, the traveltime of a spherical wave within the grid cells (green arrows in Fig. 1) near the source can be computed more accurately using the S2 operator.

For spherical waves propagating along the grid cell faces (red arrows in Fig. 1), we adopt the S4 operator. Using the *XY* plane at $z = k$ as a representative case, the partial derivatives of τ in the x and y directions in Equation (17) are discretized as

$$\begin{cases}\dfrac{\partial\tau}{\partial x}=\dfrac{\tau_{m,n,l}-\tau_{m-1,n,l}+\tau_{m,n-1,l}-\tau_{m-1,n-1,l}}{2dx},\\ \dfrac{\partial\tau}{\partial y}=\dfrac{\tau_{m,n,l}-\tau_{m,n-1,l}+\tau_{m-1,n,l}-\tau_{m-1,n-1,l}}{2dy}.\end{cases} \tag{21}$$

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For notational convenience, we define the intermediate variables τ_{px} and τ_{py} as

$$\begin{cases} \tau_{sx} = \tau_{m-1,n,l} - \tau_{m,n-1,l} + \tau_{m-1,n-1,l}, \\ \tau_{sy} = \tau_{m,n-1,l} - \tau_{m-1,n,l} + \tau_{m-1,n-1,l}. \end{cases} \tag{22}$$

The solution is

$$\begin{cases} \tau_{m,n,l} = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \\ a = \dfrac{T_0^2}{4}\left(\dfrac{1}{dx^2} + \dfrac{1}{dy^2}\right) + T_0\left(\dfrac{T_{0,x}}{dx} + \dfrac{T_{0,y}}{dy}\right) + S_0^2, \\ b = -\dfrac{T_0^2}{2}\left(\dfrac{\tau_{sx}}{dx^2} + \dfrac{\tau_{sy}}{dy^2}\right) - T_0\left(\dfrac{\tau_{sx}T_{0,x}}{dx} + \dfrac{\tau_{sy}T_{0,y}}{dy}\right), \\ c = \dfrac{T_0^2}{4}\left(\dfrac{\tau_{sx}^2}{dx^2} + \dfrac{\tau_{sy}^2}{dy^2}\right) - S_m^2. \end{cases} \tag{23}$$

For the *XZ* and *YZ* planes, $T_{m,n,l}$ can be derived in the same manner. By comparing the first-arrival traveltimes computed on the three planes, the minimum value is selected as the numerical traveltime solution at (*m*, *n*, *l*).

The R2 operator is used to compute the direct and refracted waves traveling along the grid edges (blue arrows in Fig. 1), and is written as

$$\begin{cases} T_{m,n,l} = \min(T_{rx}, T_{ry}, T_{rz}), \\ T_{rx} = T_{m-1.n.l} + dx \min(S_{m,n,l}, S_{m,n+1,l}, S_{m,n,l+1}, S_{m,n+1,l+1}), \\ T_{ry} = T_{m.n-1.l} + dy \min(S_{m,n,l}, S_{m+1,n,l}, S_{m,n,l+1}, S_{m+1,n,l+1}), \\ T_{rz} = T_{m.n.l-1} + dz \min(S_{m,n,l}, S_{m+1,n,l}, S_{m,n+1,l}, S_{m+1,n+1,l}). \end{cases} \tag{24}$$

AFDM uses Gauss–Seidel iteration with alternating sweeps in eight directions. During each sweep, to simulate all possible seismic wave propagation modes, all the local operators from Equations (11), (15), (20), (23), and (24) are used to compute the traveltimes at each node, and the minimum value is selected as the current solution. During the iterative process, updates are performed when the new value is smaller than the old value, and the iteration continues until the convergence criterion is satisfied. Additional implementation details are provided by Qiao *et al*. (2024).

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We evaluated AFDM using a constant-gradient slowness model (Fig. 2). The model contained 101 × 101 × 101 grids with a 10 m sampling interval in each spatial direction. The slowness at the surface was 0.5 s/km, and the constant gradient in the *Z*-direction was -0.1 s/km. The source was positioned at the origin of the coordinate system (0 m, 0 m, 0 m). The first-arrival traveltimes were compared with the analytical solution, and Table 1 summarizes the iteration, elapsed time, and *L*1 error for the three methods.

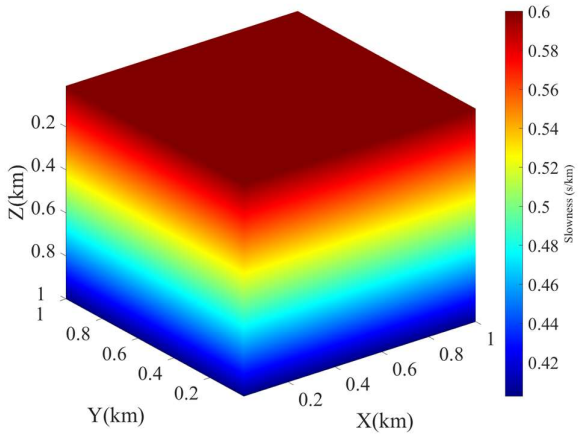


Figure 2 The constant-gradient slowness model.

Figure 3a shows the absolute error of the fast sweeping method (FSM) (Zhao 2005). It is evident that the FSM provides relatively high traveltime computation accuracy near the coordinate axes but exhibits larger errors in the diagonal directions, and these errors increase with propagation distance. The multiplicative factor decomposition method (MFDM) (Fomel *et al.* 2009) reduces source singularity and improves near-source accuracy (see Fig. 3b). However, farther from the source, the wavefronts of seismic waves progressively change from spherical to approximately planar. Owing to this transformation, the MFDM becomes less valid, leading to a decline in the computational accuracy of traveltimes, which can even fall below the performance of the FSM. The AFDM produces the smallest errors both near the source and at large offsets (Fig. 3c). Figures 3d–f show the corresponding flattened views of Figs. 3a–c for clarity. As shown in Table 1, AFDM has the smallest L1 error and the shortest runtime among the three methods. Therefore, we used the AFDM

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traveltimes as a reference for excitation time selection in the RTM.

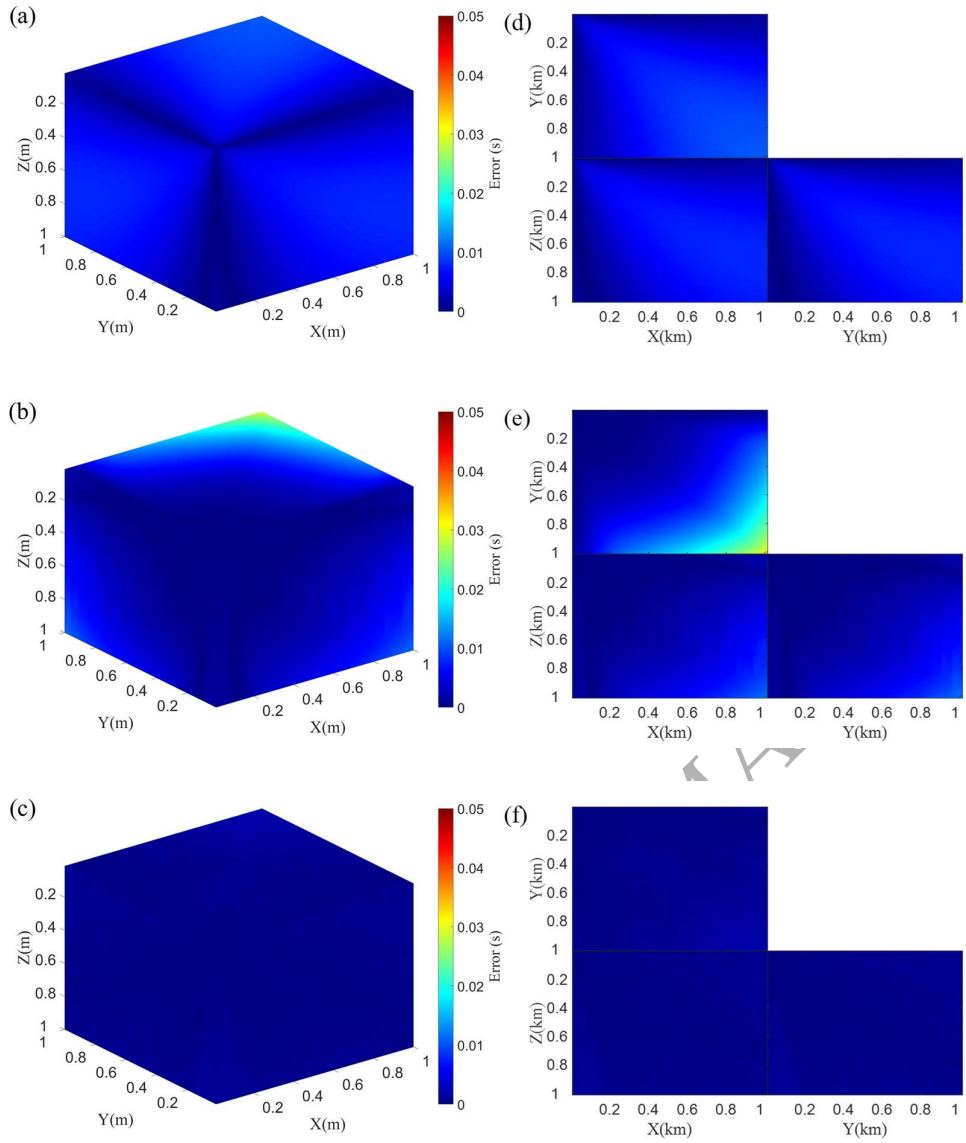


Figure 3 Absolute errors comparing the analytical solution with numerical results from (a) the FSM, (b) the MFDM, and (c) the AFDM. (d), (e), and (f) correspond to the flattened views of (a), (b), and (c), respectively.

Table 1 Comparison of iteration, time, and $L1$ error of the three methods

Methods	FSM	MFDM	AFDM
Iterations	102	4	19
Time (s)	11.42	14.47	8.11
$L1$ error (s)	1.4e-2	0.13	1.2e-2

2.3. The EAIC based on accurate first-arrival travel-time calculation

Although AFDM provides more accurate first-arrival traveltimes, it cannot be directly matched with

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the wavefront simulated by solving the wave equation using finite differences. Therefore, we apply a frequency-dependent correction given by

$$T_{\mathrm{re}} = T_{\mathrm{AFDM}} + \frac{a}{f_m}, \quad a \in [1, 1.25], \tag{25}$$

where T_{re} denotes the reference traveltimes applied to the RTM, T_{AFDM} represents the results calculated using AFDM, f_m is the dominant frequency, and a is an error factor used to correct the traveltime. In the numerical example section, we conducted a sensitivity analysis of the value range of a.

Using T_{re} as the center of the search window, we define the excitation amplitude as

$$S_{\max}(\mathbf{x}) = \max\{P_{\mathrm{s}}(\mathbf{x}; T)\}, T \in [T_{\mathrm{re}} - dt, T_{\mathrm{re}} + dt], \tag{26}$$

where max is the maximum value, T is the imaging time, and dt is the simulation time step. By concurrently storing the receiver wavefield R associated with the excitation time at every grid point, we establish a more stable excitation amplitude imaging condition (SEAIC) as

$$I_{\mathrm{SEAIC}}(\mathbf{x}; s_j) = R(\mathbf{x}; T; s_j) \cdot S_{\max}(\mathbf{x}; T; s_j), \tag{27}$$

where I_{SEAIC} is the final image generated by the SEAIC.

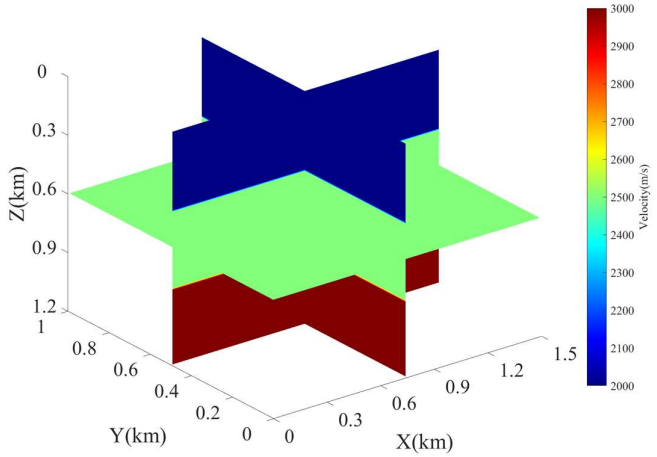
3. Applications

3.1. Simple model example

We first tested the method on a 3D simple layered model with 150 × 100 × 120 grids and a 10 m spatial sampling interval (Fig. 4). In the velocity model, the top, middle, and bottom layers were assigned the velocities of 2000, 2500, and 3000 m/s, respectively. The acquisition geometry comprised of 28 × 10 shot grids. The shot spacing was 100 m in the Y-direction and 50 m in the X-direction, and 15,000 receivers were distributed uniformly on the surface. The source was a 20 Hz Ricker wavelet, recorded for 2 s with a time step of 0.5 ms. The smoothed velocity model illustrated in Fig. 5 was used for migration.

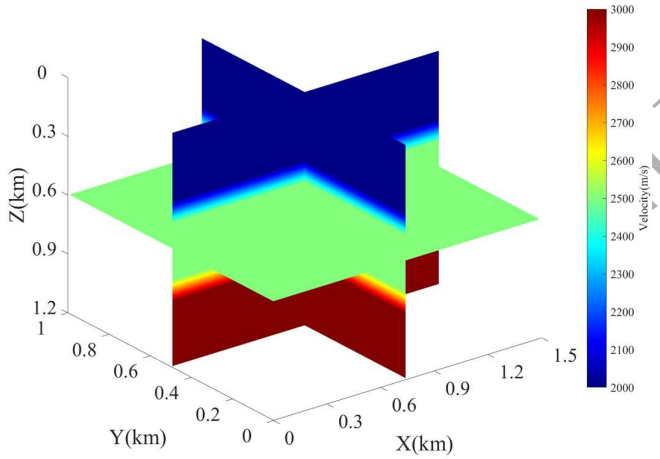
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272

273 **Figure 4** The three-layered velocity model.



274

275 **Figure 5** The smoothed three-layered velocity model.

276 Figures 6a-c present the migration images computed using the SNCCIC, EAIC, and SEAIC,
277 respectively. For clearer visualization, we display flattened views of these results in Figs. 6d-f. All
278 three imaging conditions can recover the interfaces in the simple model. To clearly demonstrate the
279 advantages of the proposed method, we extracted waveform curves from the *XZ*-plane as shown in
280 Fig. 7. The black, green, and red curves represent the SNCCIC, EAIC, and SEAIC, respectively.
281 Notably, we have marked the anomalous troughs of SNCCIC and EAIC with blue arrows, which
282 impact the resolution of the migration images, but our proposed SEAIC with the red line does not
283 exhibit this issue.

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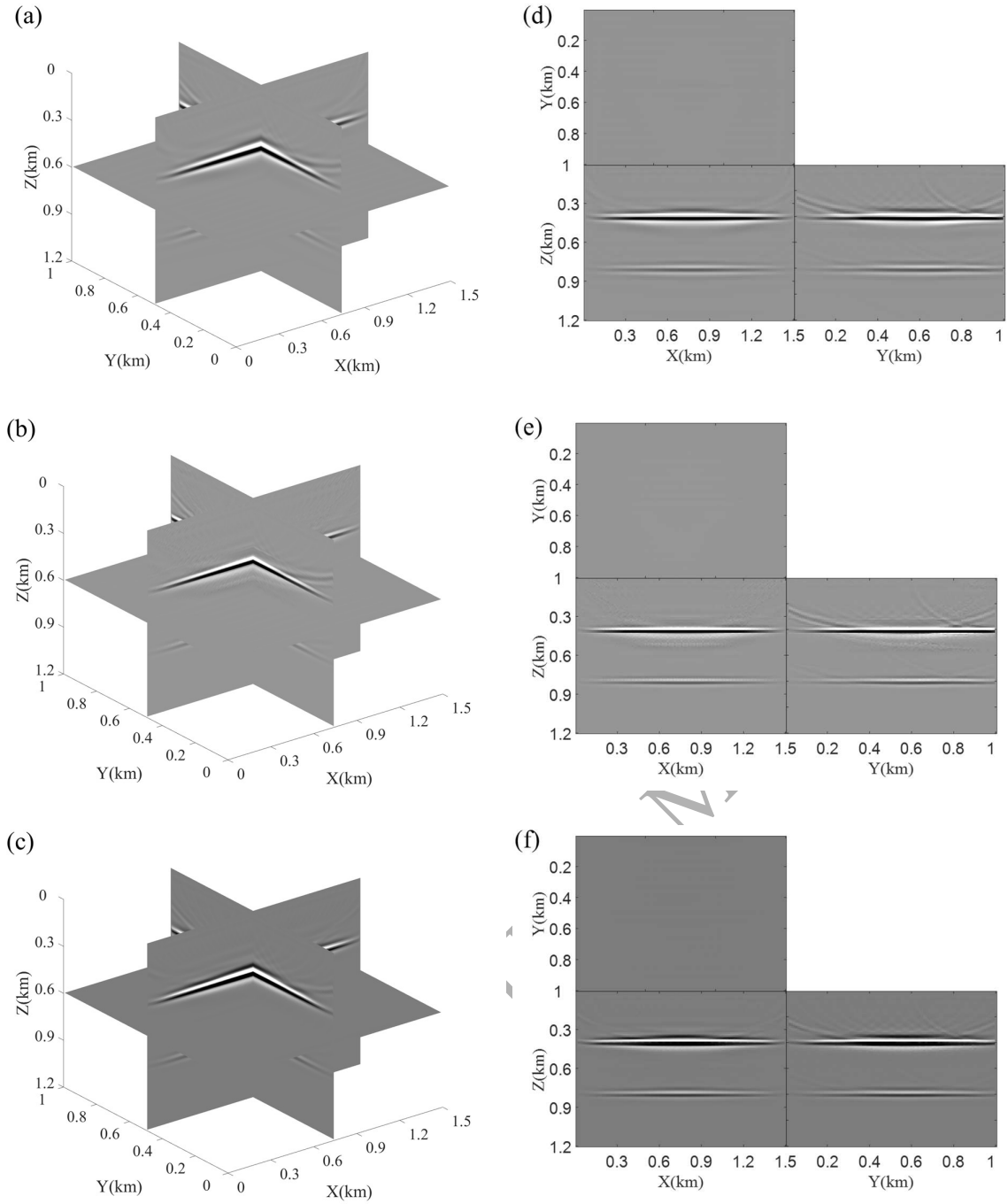
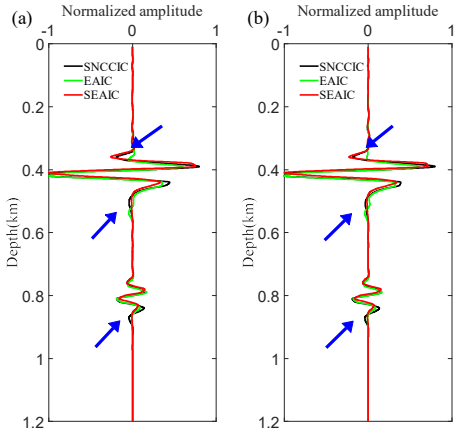


Figure 6 The migration results produced by (a) SNCCIC, (b) EAIC and (c) SEAIC. (d), (e), and (f) correspond to the flattened views of (a), (b), and (c), respectively.

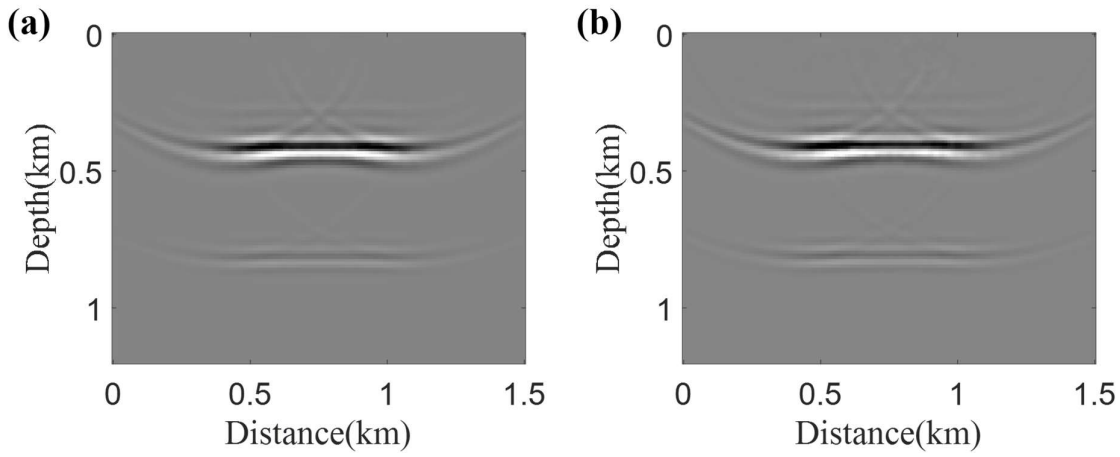
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290 **Figure 7** The waveform curves extracted from migration images using three different imaging conditions,
291 taken from the *XZ*-plane for (a) x = 500 m and (b) x = 1000 m.

292 To further discuss the effect of the error factor, we set a to 0.9, 1.0, 1.1, 1.2, and 1.3 for imaging, and
293 used the SNCCIC imaging results as references. The single-shot RTM images are shown in Fig. 8.
294 All imaging results revealed subsurface structures, but the waveform comparison in Fig. 9 revealed
295 clear differences. Despite yielding slightly weaker amplitudes, the values of a within the range
296 defined by Equation (25) matched the reference waveform more closely. In contrast, when a = 0.9 or
297 a = 1.3, the curves have significant deviations, which prevent the image from recovering the true
298 spatial positions and degrade the imaging quality.



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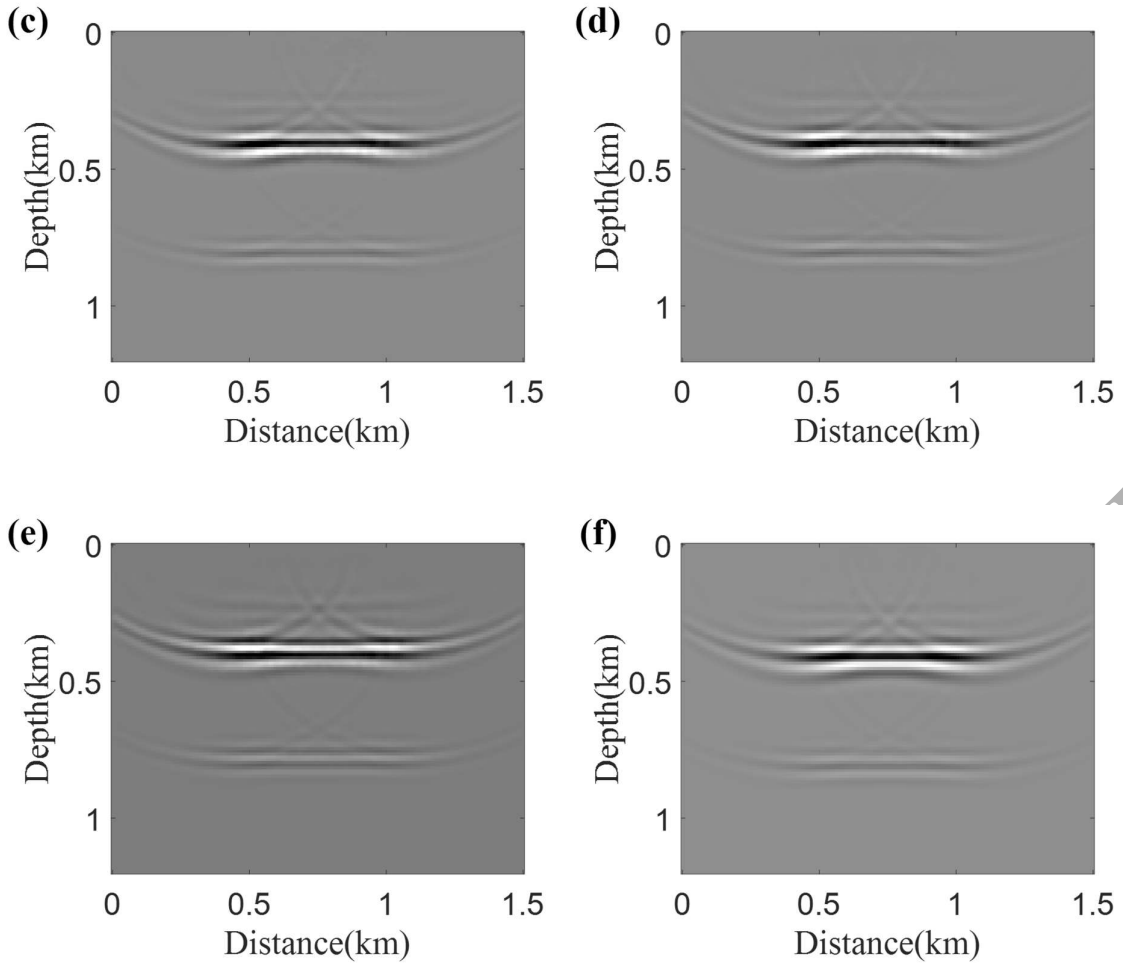


Figure 8 Results of single-shot RTM imaging using the SEAIC error factors of (a) a = 0.9, (b) a = 1.0, (c) a = 1.1, (d) a = 1.2, (e) a = 1.3 and (f) SNCCIC as a reference.

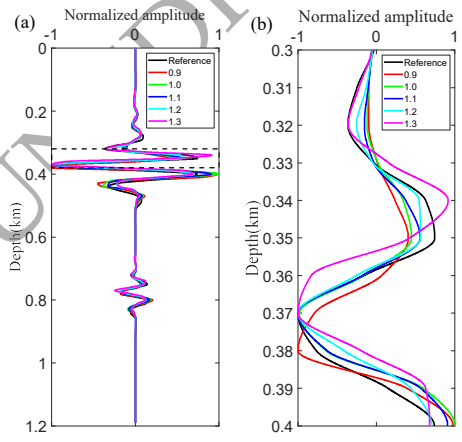


Figure 9 (a) Comparison of waveform curves of SEAIC single-shot migration images using different error factors, with SNCCIC as a reference. (b) Detailed view of the black dashed box location in (a).

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3.2. Salt model example

The model had a grid size of 696 × 120 × 185, with a spatial sampling interval of 10 m (Fig. 10). A 20 Hz Ricker wavelet was used as the source. The recording duration was 3 s, and the time sampling interval was 0.5 ms. A total of 45 × 10 sources and 83,520 receivers were uniformly distributed on the surface. The migration velocity model is shown in Fig. 11.

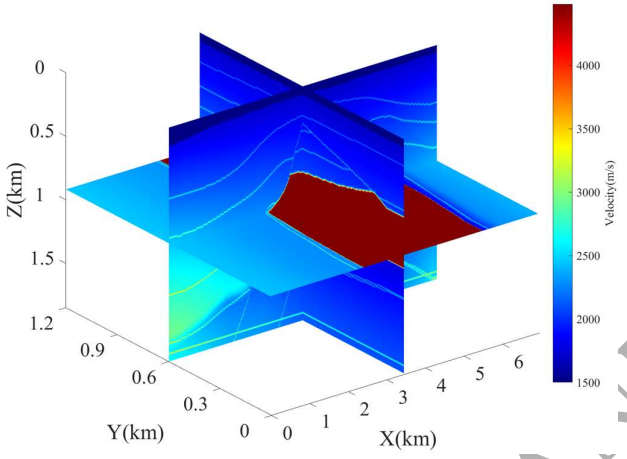


Figure 10 The 3D Salt velocity model.

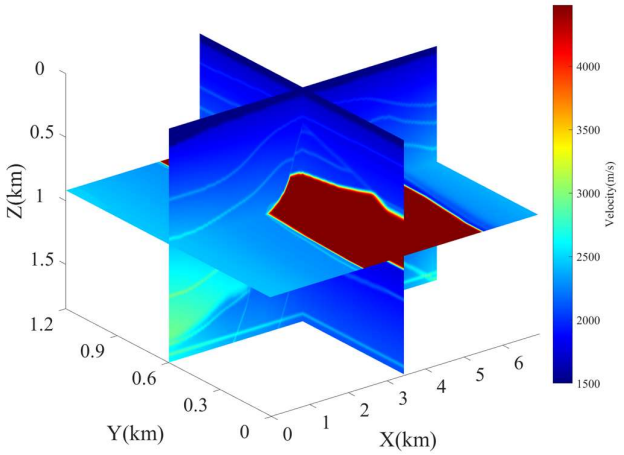


Figure 11 The smoothed 3D Salt velocity model.

Figures 12a and 12b show the excitation time and excitation amplitude maps obtained using the EAIC for a source positioned at (150 m, 150 m, 0 m). It can be observed that both fields exhibit significant chaotic characteristics, and this chaos exhibits a distinct spatiotemporal correspondence (as indicated by the black arrows). This instability directly affects the clarity of the migration results. In contrast, the excitation time computed by the SEAIC using AFDM (Fig. 12c) exhibits good

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continuity and stability, ensuring the quality of the calculated excitation amplitude (Fig. 12d). A comparison between Fig. 12b and Fig. 12d also clearly shows that the excitation amplitude obtained using the SEAIC reflects the subsurface structural features more clearly, whereas the amplitude from the EAIC exhibits significant distortion.

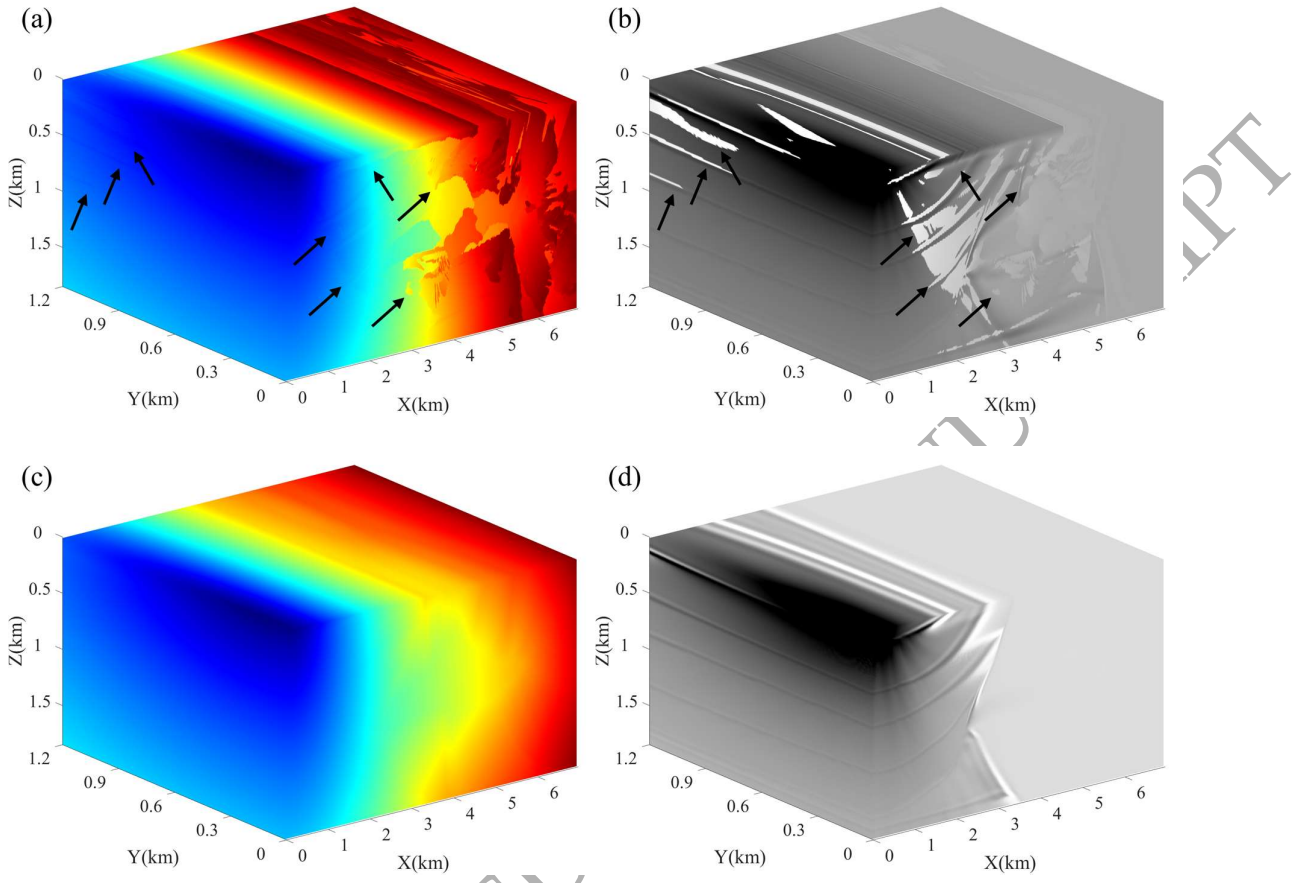


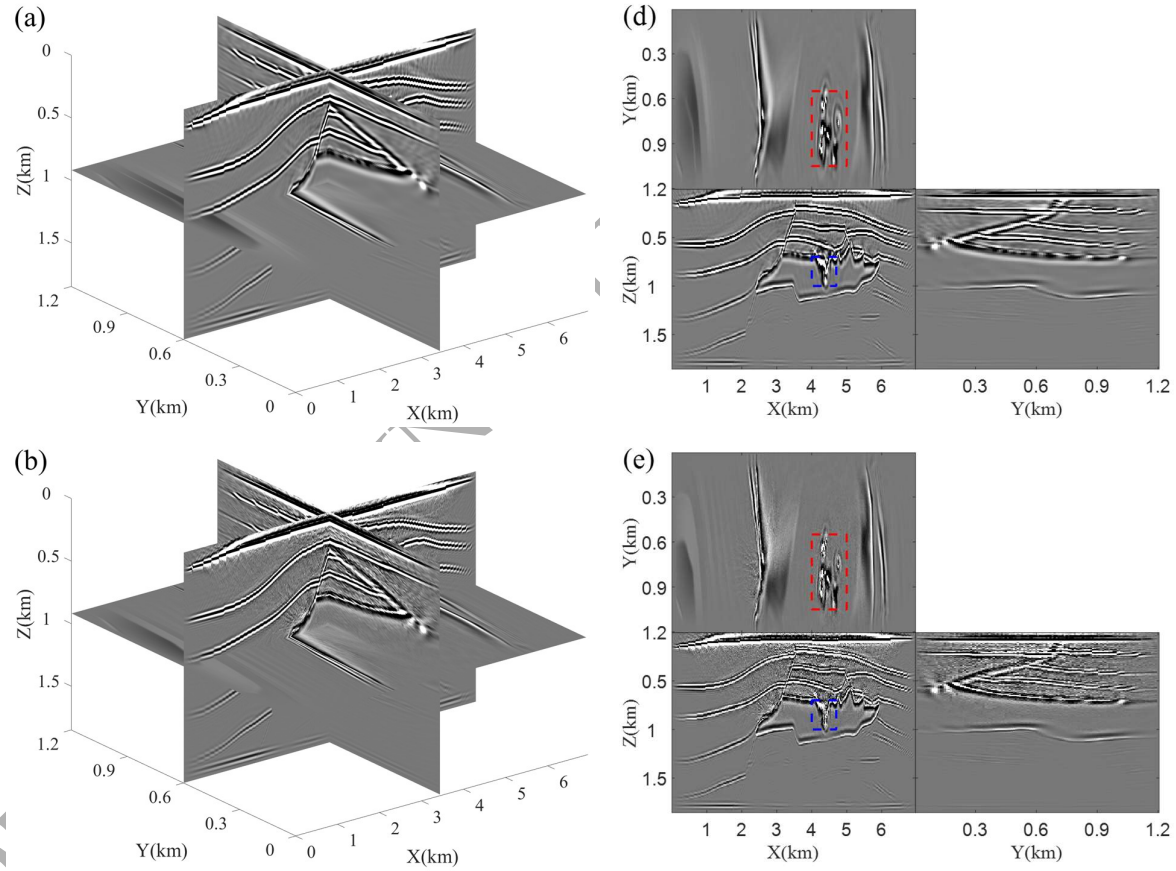
Figure 12 The maps of (a) the excitation time map and (b) the excitation amplitude map of EAIC, and (c) and (d) are similar to (a) and (b), but correspond to SEAIC instead. The source is placed at (150 m, 150 m, 0 m).

Figures 13a–c show the RTM images obtained with SNCCIC, EAIC, and our proposed SEAIC, respectively. Figure 14 presents the true reflectivity model, which serves as a reference for evaluating the imaging results. For the reflectors above the salt body, all three imaging methods reconstruct the structural geometry and spatial positioning. However, in the EAIC method, the imaging time is determined by the maximum excitation amplitude. Under complex wavefield conditions, this estimation is often inaccurate (as shown in Fig. 12a and Fig. 12b), resulting in significant noise in the images and consequently lower overall resolution.

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Near the top of the salt body, strong velocity contrasts and complex geometries lead to highly complex wave propagation paths. In addition to the primary reflections, the wavefield contains considerable contributions from diffractions, multiples, and turning waves. Although the imaging is still dominated by primary reflections, the influence of these complex wave components cannot be neglected. Because SNCCIC utilizes wavefield information over all time steps, it inevitably incorporates these interfering waves, leading to noticeable migration artifacts in localized areas (as indicated by the dashed boxes in the figures), which degrade the image quality. In contrast, SEAIC constrains the imaging time by using more accurate first-arrival traveltimes, allowing the imaging process to focus primarily on the first-arriving energy. This effectively suppresses the influence of complex wave interference. Consequently, SEAIC produces clearer imaging results in structurally complex regions. The SEAIC also recovers reflectors beneath the salt body, although their amplitudes are slightly weaker because they utilize wavefield information from only one sample.



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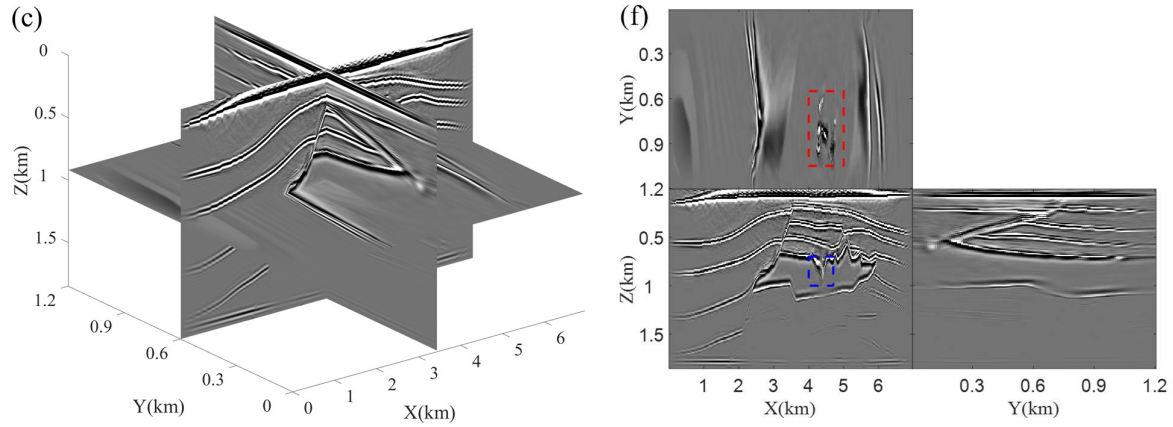


Figure 13 The migration results produced by (a) SNCCIC, (b) EAIC and (c) SEAIC. (d), (e), and (f) correspond to the flattened views of (a), (b), and (c), respectively.

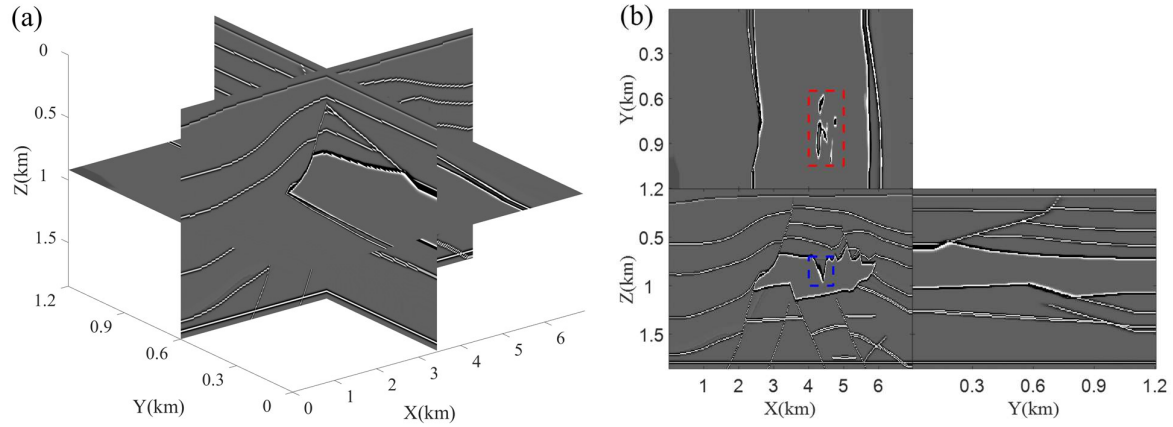


Figure 14 (a) True reflectivity model and (b) flattened view.

To quantify the image accuracy, we defined the maximum absolute value error (E_{mav}) and root-mean-square error (E_{rms}) as

$$E_{\mathrm{mav}} = \max\left|\frac{I_{i,j,k} - I^{\mathrm{r}}_{i,j,k}}{I^{\mathrm{r}}_{\max}}\right|, \tag{28}$$

and

$$E_{\mathrm{rms}} = \sqrt{\frac{1}{N_x N_y N_z}\sum_{i=1}^{N_x}\sum_{j=1}^{N_y}\sum_{k=1}^{N_z}\left(\frac{I_{i,j,k} - I^{\mathrm{r}}_{i,j,k}}{I^{\mathrm{r}}_{\max}}\right)^2}, \tag{29}$$

where I represents the migration images, I^{r} represents the reflectivity model, $I^{\mathrm{r}}_{\max}$ represents the maximum value of the reflectivity model, and N_x, N_y, and N_z denote the grid dimensions. Table 2 presents a comparison of the errors under the three imaging conditions relative to the reflectivity

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model. We can see that, among the three different imaging conditions, our proposed SEAIC exhibits the smallest E_{mav}. Furthermore, its E_{rms} is comparable to that of the SNCCIC and significantly outperforms that of the EAIC.

Table 2 Comparison of E_{mav} and E_{rms} of the three imaging results.

Methods	E_{mav}	E_{rms}
SNCCIC	0.99	3.53e-2
EAIC	0.98	5.64e-2
SEAIC	0.97	3.82e-2

Figure 15a compares the disk storage costs under the three imaging conditions. The SNCCIC requires storage of wavefield data at all time steps, resulting in extremely large storage demands. In contrast, SEAIC only requires additional storage for the travel-time information computed by the AFDM on top of the EAIC, and thus maintains a relatively low overall storage cost.

In terms of computational efficiency (Fig. 15b), all three imaging methods were implemented using the MPI+GPU parallel framework. SNCCIC involves frequent data transfers among the GPU, CPU, and hard disk, resulting in substantial I/O overhead and, consequently, very low computational efficiency, with a total elapsed time of 154.34 h. In contrast, both EAIC and SEAIC exhibited significantly higher computational efficiencies than SNCCIC. Considering only the RTM process itself, SEAIC requires 11.51 h, which is slightly faster than EAIC (11.79 h). Although AFDM introduces some additional computational cost, its elapsed time of only 0.75 h is entirely acceptable compared with the overall RTM computational time.

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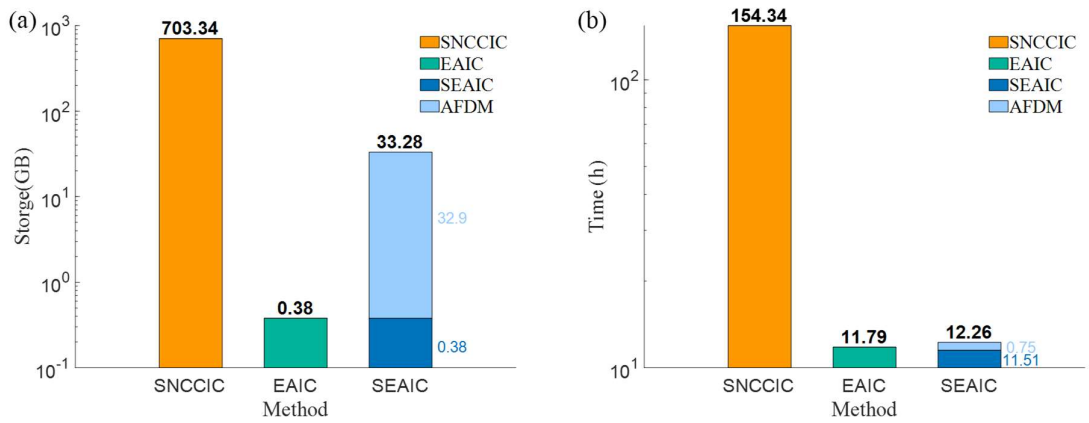


Figure 15 Comparisons between SNCCIC, EAIC, and SEAIC in terms of (a) storage costs and (b) elapsed time.

4. Discussion

The methodological advantage of SEAIC lies in introducing a physically defined traveltime constraint into the economical sampling strategy of excitation-amplitude imaging. In conventional EAIC, the imaging time is determined by the largest source-wavefield amplitude, whose occurrence depends on geometrical spreading, focusing, transmission, and interference between arrivals. In heterogeneous media, neighboring grid points may therefore select different propagation branches, disrupting the consistency of source–receiver wavefield matching. First-arrival guidance reduces this ambiguity by linking excitation-time selection to the propagation kinematics of the migration velocity model. Building on earlier traveltime-guided imaging (Gu *et al*. 2015; Wang *et al*. 2025), the present method implements this principle in 3D RTM. Its main contribution is a way to constrain event selection while retaining the limited wavefield storage of excitation-based imaging, thereby supporting coherent structural imaging without requiring the complete source-wavefield history.

The 3D AFDM provides the kinematic basis for this constraint. Its local operators represent spherical and plane wavefronts, together with direct or refracted propagation along grid edges, and are combined through causal updates and multidirectional sweeping (Qiao *et al*. 2024). This formulation addresses near-source wavefront curvature and propagation through heterogeneous cells without imposing a single local wavefront approximation throughout the model. For EAIC, these

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properties matter because traveltime errors directly affect which source-wavefield sample is paired with the back-propagated receiver wavefield. The frequency-dependent time correction and local amplitude search then connect the eikonal arrival to the finite-bandwidth source pulse. The excitation amplitude remains an output of acoustic wave-equation modeling, so the eikonal solver supplies timing information rather than replacing wavefield extrapolation. This separation allows SEAIC to combine a consistent traveltime reference with the wave-propagation physics already represented by acoustic RTM.

The storage advantage follows from the imaging condition itself. For each shot, SEAIC retains spatial fields describing the selected excitation times and amplitudes, together with the required traveltime information, instead of a sequence of source-wavefield snapshots. The retained source information therefore scales with the number of spatial grid points rather than their product with the number of stored time levels. During receiver back-propagation, imaging contributions are evaluated at the selected times, avoiding repeated retrieval of the complete source-wavefield history. This reduces disk traffic and transfers between storage and computational memory, which are particularly demanding for wavefields. Unlike approaches that recover the source history through recomputation, SEAIC directly retains the information required by its imaging condition. Both wavefields still require propagation through the recording interval, and the eikonal calculation introduces additional work; the principal savings concern wavefield retention, data access, and temporal accumulation during imaging.

These advantages come at the cost of incomplete wavefield sampling. A single first-arrival-guided sample cannot represent all propagation branches, so later arrivals that illuminate structures beneath salt or strong velocity contrasts may be excluded; coherent timing alone therefore does not ensure complete illumination or true-amplitude imaging. Limited temporal averaging also makes SEAIC sensitive to receiver-wavefield noise. To examine this limitation, Gaussian noise was added to the seismic records at an S/N of 10 dB using Seismic Unix (Cohen & Stockwell 2008), as shown

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in Fig. 16. Figure 17 compares SNCCIC, SEAIC, and local cross-correlation imaging condition (LCCIC) in Figs. 17a–c, respectively, with their corresponding flattened views in Figs. 17d–f. The stronger background noise in the SEAIC image illustrates the vulnerability of single-sample imaging, whereas LCCIC suppresses more of the noise by correlating wavefields within a finite window centered on the AFDM first-arrival traveltimes (Zhang *et al.* 2022; Gao *et al.* 2025; Liu *et al.* 2025). This extension retains nearby arrivals and provides temporal averaging without storing the complete source-wavefield history. The LCCIC computation time of 22.49 h exceeds that of SEAIC but remains below that of SNCCIC, reflecting the trade-off between temporal coverage and computational cost. Window length must nevertheless balance useful arrivals, interference, and storage requirements, and arrivals outside the window remain unavailable. Future work should assess sensitivity to source-wavelet phase and velocity errors and establish applicability to field data and more complex propagation physics.

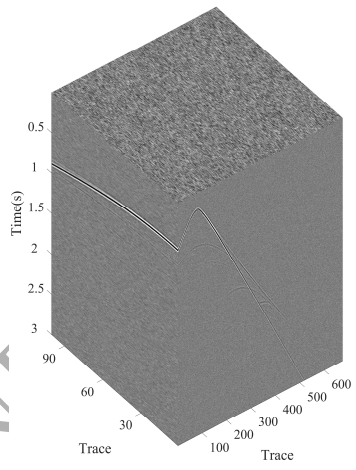


Figure 16 Seismic records after adding random Gaussian noise, with S/N = 10 dB.

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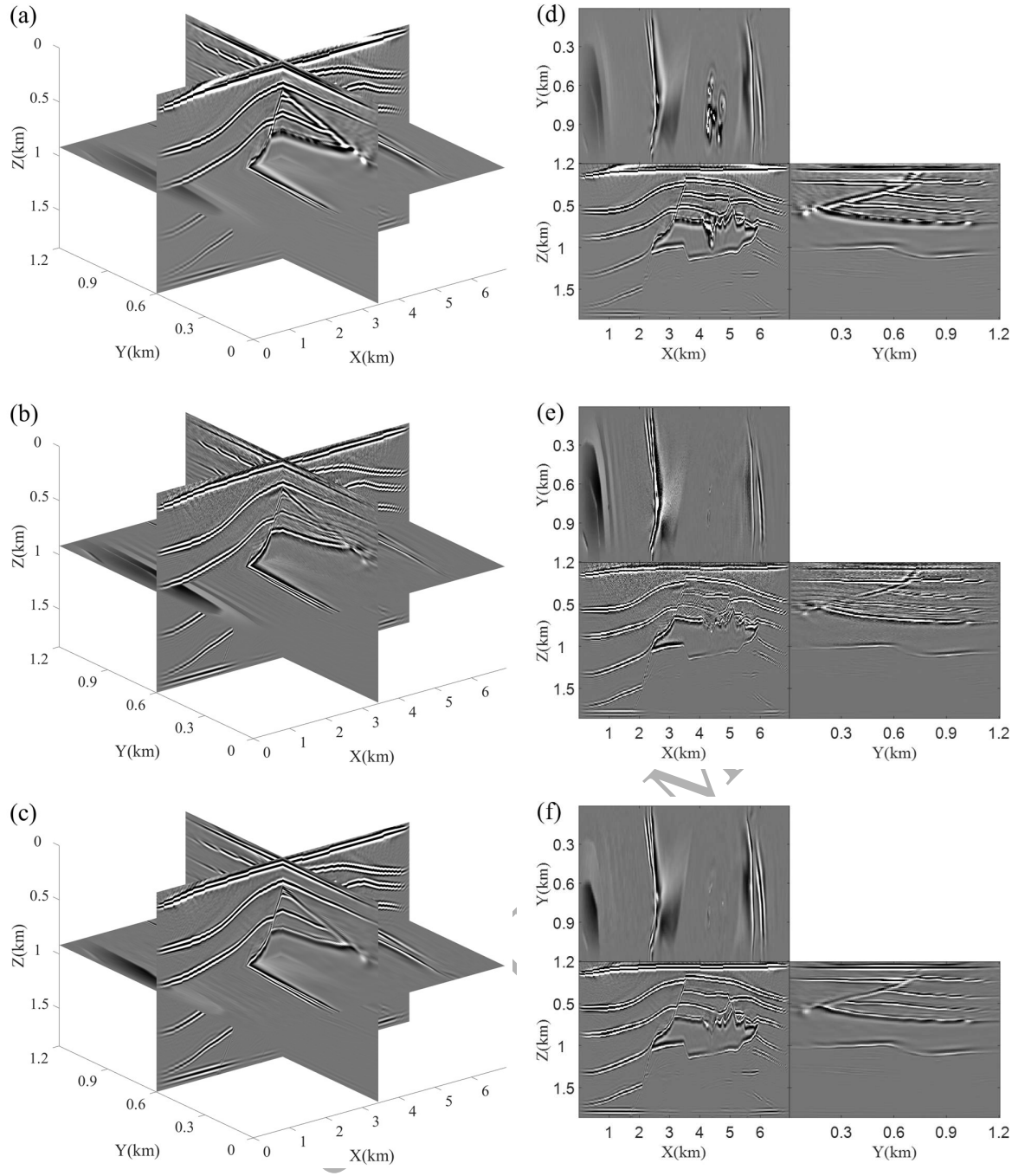


Figure 17 The migration results produced by (a) SNCCIC, (b) SEAIC and (c) LCCIC. (d), (e), and (f) correspond to the flattened views of (a), (b), and (c), respectively.

5. Conclusions

In this paper, we developed SEAIC for 3D acoustic RTM, in which first-arrival traveltimes computed from the 3D eikonal equation using AFDM guide excitation-time selection. By linking the imaging sample to the kinematics of the first-arriving source wavefield, SEAIC reduces spatial

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inconsistencies in excitation times and amplitudes associated with conventional peak-amplitude picking, thereby improving the coherence of migrated reflection events in heterogeneous media. Numerical experiments on layered and Salt models demonstrate improved reflector continuity and reduced migration artifacts compared with conventional EAIC. These imaging improvements are achieved without storing and retrieving the complete source-wavefield history, substantially reducing storage requirements and computational costs. Thus, the SEAIC method offers an efficient approach to 3D seismic depth imaging of complex subsurface structures, with potential applications in exploration settings where storage and computational resources constrain migration workflows.

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Conflict of interest statement

None declared.

Data Availability

Data associated with this research are available and can be obtained by contacting the corresponding authors.

References

Baysal E, Kosloff DD, Sherwood JW. 1983. Reverse time migration. *Geophysics,* **48**:1514-24. http://dx.doi.org/10.1190/1.1441434

Chang H, VanDyke JP, Solano M, McMechan GA, Epili D. 1998. 3-D prestack Kirchhoff depth migration: From prototype to production in a massively parallel processor environment. *Geophysics,* **63**:546-56. http://dx.doi.org/10.1190/1.1444355

Chattopadhyay S, McMechan GA. 2008. Imaging conditions for prestack reverse-time migration. *Geophysics,* **73**:S81-9. http://dx.doi.org/10.1190/1.2903822

Chen GT, Wang ZL. 2018. A checkpoint-assisted interpolation algorithm of wave field reconstruction and prestack reverse time migration. *Chinese J Geophys,* **61**:3334-45. http://dx.doi.org/10.6038/cjg2018L0504

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Downloaded from academic.oup.com/jge/advance-article/doi/10.1093/jge/gxag124/8835800 by guest on 04 November 2026

Claerbout JF. 1971. Toward a unified theory of reflector mapping. *Geophysics,* **36**:467-81. http://dx.doi.org/10.1190/1.1440185

Clapp RG. 2009. Reverse time migration with random boundaries. SEG Expanded Abstracts: 2809-13. http://dx.doi.org/10.1190/1.3255432

Cohen J, Stockwell J. 2008. CWP/SU: seismic unix release No. 41: an open source software package for seismic research and processing. *Center for Wave Phenomena, Colorado School of Mines.*

Du QZ, Zhang MQ, Gong XF, Chen XR. 2014. Polarity-consistent excitation amplitude imaging condition for elastic reverse time migration. *J Geophys Eng,* **12**:33-44. http://dx.doi.org/10.1088/1742-2132/12/1/33

Feng B, Wang HZ. 2012. Reverse time migration with source wavefield reconstruction strategy. *J Geophys Eng,* **9**:69-74. http://dx.doi.org/10.1088/1742-2132/9/1/008

Fomel S, Luo ST, Zhao HK. 2009. Fast sweeping method for the factored eikonal equation. *J Comput Phys,* **228**:6440-55. http://dx.doi.org/10.1016/j.jcp.2009.05.029

Gao LJ, Li ZJ, Li HY, Yang W, Zhang Q. 2025. Reverse time migration based on local Nyquist cross-correlation imaging condition with accurate first-arrival traveltimes correction. *Front Earth Sci,* **13**:1605436. http://dx.doi.org/10.3389/feart.2025.1605436

Gazdag J, Sguazzero P. 1984. Migration of seismic data by phase shift plus interpolation. *Geophysics,* **49**:124-31. http://dx.doi.org/10.1190/1.1441643

Gray SH. 2005. Gaussian beam migration of common-shot records. *Geophysics,* **70**:S71-7. http://dx.doi.org/10.1190/1.1988186

Griewank A. 1992. Achieving logarithmic growth of temporal and spatial complexity in reverse automatic differentiation. *Optim Method Softw,* **1**:35-54. http://dx.doi.org/10.1080/10556789208805505

Gu BL, Liu YS, Ma XN, Li ZY, Liang GH. 2015. A modified excitation amplitude imaging condition for prestack reverse time migration. *Explor Geophys,* **46**:359-70. http://dx.doi.org/10.1071/eg14039

Hill NR. 2001. Prestack Gaussian-beam depth migration. *Geophysics,* **66**:1240-50. http://dx.doi.org/10.1190/1.1487071

Hu H, Xia KY, Hilterman F, Zhang Y. 2020. Amplitude-Versus-Angle Analysis of Local Angle-Domain Common Image Gathers With Prestack Gaussian-Beam Migration of Seismic Data. *IEEE Trans Geosci Remote Sensing,* **58**:5969-75. http://dx.doi.org/10.1109/tgrs.2020.2973654

Kaelin B, Guitton A. 2006. Imaging condition for reverse time migration. SEG Expanded Abstracts: 2594-98. http://dx.doi.org/10.1190/1.2370059

Li J, Du GN, Qin DW, Yin WS, Tan J, Liu ZL, Song P. 2024. Excitation-time imaging condition reverse-time migration based on a physics-informed neural network traveltime calculation with wavefield decomposition using an optical flow vector. *J Geophys Eng,* **21**:200-20. http://dx.doi.org/10.1093/jge/gxad106

Liu W, Shi Y, Wang N, Li SL. 2025. Efficient source-independent Q-compensated least-squares reverse time migration with LNCC imaging condition. *J Geophys Eng,* **22**:428-48. http://dx.doi.org/10.1093/jge/gxaf016

Loewenthal D, Mufti IR. 1983. Reversed time migration in spatial frequency domain. *Geophysics,* **48**:627-35. http://dx.doi.org/10.1190/1.1441493

Mulder WA, Plessix RE. 2004. A comparison between one-way and two-way wave-equation migration. *Geophysics,* **69**:1491-504. http://dx.doi.org/10.1190/1.1836822

Nguyen BD, McMechan GA. 2013. Excitation amplitude imaging condition for prestack reverse-time migration. *Geophysics,* **78**:S37-46. http://dx.doi.org/10.1190/geo2012-0079.1

Nguyen BD, McMechan GA. 2015. Five ways to avoid storing source wavefield snapshots in 2D elastic prestack reverse time migration. *Geophysics,* **80**:S1-18. http://dx.doi.org/10.1190/geo2014-0014.1

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Noble M, Gesret A, Belayouni N. 2014. Accurate 3-D finite difference computation of traveltimes in strongly heterogeneous media. *Geophys J Int,* **199**:1572-85. http://dx.doi.org/10.1093/gji/ggu358

Pasasa L, Wenzel F, Zhao P. 1998. Prestack Kirchhoff depth migration of shallow seismic data. *Geophysics,* **63**:1241-47. http://dx.doi.org/10.1190/1.1444425

Qiao BP, Pan ZQ, Huang WC, Cao CY. 2021. An adaptive finite-difference method for accurate simulation of first-arrival traveltimes in heterogeneous media. *Appl Math Comput,* **394**:125792. http://dx.doi.org/10.1016/j.amc.2020.125792

Qiao BP, Li QQ, He WG, Zhao D, Wu QB. 2024. An adaptive finite-difference method for seismic traveltime modeling based on 3D eikonal equation. *Pet Sci,* **21**:195-205. http://dx.doi.org/10.1016/j.petsci.2023.09.013

Schneider WA. 1978. Integral formulation for migration in two and three dimensions. *Geophysics,* **43**:49-76. http://dx.doi.org/10.1190/1.1440828

Sethian JA. 1996. A fast marching level set method for monotonically advancing fronts. *Proc Natl Acad Sci U S A,* **93**:1591-95. http://dx.doi.org/10.1073/pnas.93.4.1591

Symes WW. 2007. Reverse time migration with optimal checkpointing. *Geophysics,* **72**:SM213-21. http://dx.doi.org/10.1190/1.2742686

Wang J, Li QQ, Qiao BP, Qi JX. 2025. Reverse Time Migration Using Excitation Amplitude Imaging Condition Based on Accurate First-Arrival Traveltimes Calculation. *IEEE Trans Geosci Remote Sensing,* **63**:1-11. http://dx.doi.org/10.1109/tgrs.2025.3527144

Yue YB, Liu YJ, Ye YF, Wo YK, Qian ZP. 2023. Least-Squares Kirchhoff Depth Migration With Fast Point-Spread-Function Computation. *IEEE Trans Geosci Remote Sensing,* **61**:1-10. http://dx.doi.org/10.1109/tgrs.2023.3275811

Zhang JM, Dong LG, Huang C. 2024. A shortest-path-aided fast-sweeping method to improve the accuracy of traveltime calculation in vertically transverse isotropic media. *Geophys Prospect,* **72**:2761-71. http://dx.doi.org/10.1111/1365-2478.13537

Zhang MK, Zhou H, Chen HM, Jiang SQ, Jiang CT. 2022. Reverse-Time Migration Using Local Nyquist Cross-Correlation Imaging Condition. *IEEE Trans Geosci Remote Sensing,* **60**:1-14. http://dx.doi.org/10.1109/tgrs.2022.3168582

Zhang QC, Mao WJ, Chen YK. 2019. Attenuating Crosstalk Noise of Simultaneous-Source Least-Squares Reverse Time Migration With GPU-Based Excitation Amplitude Imaging Condition. *IEEE Trans Geosci Remote Sensing,* **57**:587-97. http://dx.doi.org/10.1109/tgrs.2018.2858850

Zhang Y, Zhang GQ, Bleistein N. 2005. Theory of true-amplitude one-way wave equations and true-amplitude common-shot migration. *Geophysics,* **70**:E1-10. http://dx.doi.org/10.1190/1.1988182

Zhang YT, Zhao HK, Qian JL. 2006. High order fast sweeping methods for static Hamilton–Jacobi equations. *J Sci Comput,* **29**:25-56. http://dx.doi.org/10.1007/s10915-005-9014-3

Zhao HK. 2005. A fast sweeping method for eikonal equations. *Math Comput,* **74**:603-27. http://dx.doi.org/10.1090/S0025-5718-04-01678-3

Zhou HW, Hu H, Zou ZH, Wo YK, Youn O. 2018. Reverse time migration: A prospect of seismic imaging methodology. *Earth-Sci Rev,* **179**:207-27. http://dx.doi.org/10.1016/j.earscirev.2018.02.008

Zhu TF, Gray SH, Wang DL. 2007. Prestack Gaussian-beam depth migration in anisotropic media. *Geophysics,* **72**:S133-8. http://dx.doi.org/10.1190/1.2711423

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